



Mark Beyer Stjerne

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[Portfolio](#)

[GitHub](#)

[LinkedIn](#)

Astrophysicist with experience handling optical data across wavelengths from observation to data reduction, bringing a multicultural perspective and a drive to contribute to real missions.

EDUCATION

Master of Science in Physics (GPA: 10.0/12)

University of Copenhagen

Sep 2024–Present

Copenhagen, Denmark

Relevant projects/coursework (☆ = Ongoing):

- ☆ **Master's thesis.** Developing Python pipeline for processing and analysing large-scale observational datasets, implementing data quality checks, statistical modelling, and signal detection.
- ☆ **Stellar Occultations of 3I/ATLAS.** Ongoing observations of an interstellar comet as it occults bright stars to examine chemical signatures in the back-illuminated coma.
- ☆ **Niels Bohr Interferometric Teaching Telescopes (NBITT).** Contributing to the construction of a pair of radio telescopes for teaching, contributing with changes to tracking software.
- **Course: Advanced Methods in Applied Statistics.** Implemented Gaussian mixture models and two-dimensional KS testing on exoplanet population data to evaluate underlying distributional assumptions.
- **Course: Astronomical Data Processing.** Writing Python-based reduction and analysis pipelines to raw imaging and spectroscopic data; performed data reduction, uncertainty propagation, and wavelength calibration to produce usable spectra and images.

Bachelor of Science in Physics (GPA: 74.8/100)

University of New South Wales

Feb 2019–Dec 2021

Sydney, Australia

Relevant projects/coursework:

- **Undergraduate Research Internship.** Analysed GALAH survey catalog data to examine lithium depletion in red-clump stars; applied data processing and visualisation in Python, presenting my results to a research group.

SKILLS & MORE

Languages/programs: Python, $\LaTeX$, Excel, PowerBI, Git, SQL

Python libraries: NumPy, Matplotlib, Pandas, Seaborn, SciPy, Astropy

Languages (CEFR): English (C2), Danish (C1), French (A2/B1), Japanese (A2), German (A1/A2)

Citizenship: Denmark, Australia

Hobbies/interests: Astronomy, ice skating, playing piano, languages and linguistics

OTHER EXPERIENCE

Nordic Optical Telescope (NOT) Summer School 2025 [Link]

Roque del los Muchachos Observatory

Aug 2025 (3 weeks)

La Palma, Canary Islands

Selected to complete nights of observing using the 2.56m NOT (ALFOSC imaging/spectroscopy and FIES echelle spectroscopy); carried out reduction and analysis of data, and presented findings at the school's closing mini-conference.

EMPLOYMENT

Claims Handler and Data Analyst

DHL Insurance and Risk Management

Copenhagen, Denmark

Sep 2022–Present

Casual Teaching Assistant

University of New South Wales, School of Physics

Delivered Online

Jan–May 2022